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Master Thesis

"Measuring the Shape of Science: Morphological Indicators and Evolution of Research Fields"

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SUMMARY

Scientific fields are often described through labels, topics, citation links, or positions on maps. These views are useful, but they do not directly measure how the papers of a field are arranged. This thesis treats fields as distributions of embedded scientific documents and asks what can be learned by measuring their structural shape, temporal evolution, similarity, and relations.

The empirical corpus is built from OpenAlex title–abstract records covering 2000–2024. It contains 2,378,036 text-eligible works across 252 subfields, with a row-aligned SPECTER2 analysis subset of 2,344,927 embedded works across 241 analysis subfields. Quantitative metrics are computed in the original 768-dimensional SPECTER2 space; PCA and UMAP are used only for display.

The framework defines an eight-metric structural morphology profile that covers dispersion, local density, hubness, and spectral structure. Early–late centroid drift is retained separately as a measure of net semantic displacement. The results show that broad domains have tendencies but do not determine field shape. Physical Sciences are generally more dispersed and spectrally extended, while Health Sciences are more locally dense and hub-concentrated. At field level, 17 of 26 fields have their nearest morphological neighbor outside their own OpenAlex domain. Mean field-pair morphological distance rises from 1.44 in 2000–2004 to 1.81 in 2020–2024, indicating average differentiation alongside selective convergence between specific field pairs. Static morphology is weakly clustered, whereas temporal trajectories form clearer dynamic typologies.

The contribution is a measurement framework for comparing fields as distributions rather than only as labels, centroids, or visual regions. It shows why no single map is enough to describe scientific structure and gives field comparison a clearer object of analysis.

Keywords: science mapping; scientometrics; scientific document embeddings; SPECTER2; OpenAlex; embedding-space morphology; semantic space; temporal evolution; centroid drift; research fields

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1. INTRODUCTION

1.1. Problem and Contribution

Scientific fields are usually described by names, topics, citation links, publication counts, or positions on maps. These views are useful, but they do not directly measure how the papers of a field are arranged. Two fields can belong to different OpenAlex domains and still have similarly shaped paper distributions; two fields can be close in semantic location and differ sharply in dispersion, local density, hubness, or spectral structure.

This thesis treats scientific fields as distributions of embedded documents. I use OpenAlex title–abstract records, represent them with SPECTER2 document embeddings, and measure field morphology in the original 768-dimensional embedding space. By embedding-space morphology I mean the measurable structural shape of a field's paper distribution: its spread around a center, its local neighborhoods, its hub concentration, and its principal directions of variation.

The active morphology profile has eight structural metrics. Temporal evolution is measured by recomputing those same eight metrics in five-year windows. Net semantic displacement is handled separately through early–late centroid drift: it describes how far a subfield's semantic center moves, not the structural shape used in static profiles, PCA, clustering, or morphological distances.

The contribution is a reproducible framework for separating five related but distinct objects of analysis: labels, semantic locations, static shapes, temporal trajectories, and relations between fields. The thesis does not argue that morphology replaces taxonomies, citation maps, topic models, or visual maps. It argues that those tools leave a measurable layer underdescribed: the internal arrangement of papers once they have shared document coordinates.

1.2. Research Questions

The empirical analysis is organized around three research questions:

1. How can the internal structure of scientific fields be measured using scientific document embeddings?

2. How does the internal structure of scientific fields vary across disciplines and evolve over time?

3. Which scientific fields converge or diverge in their structural profiles over time?

These questions are descriptive rather than causal. A dispersed field is not better than a compact one, a dynamic field is not necessarily more important, and a low morphological distance does not mean that two fields study the same topic. The goal is to define and test a measurement language for field shape.

1.3. Roadmap

Chapter 2 positions the thesis in science mapping and scientific document representation. Chapter 3 describes the corpus, SPECTER2 representation, structural morphology metrics, temporal windows, and workflow. Chapter 4 compares static morphology across domains, fields, and subfields; Chapter 5 examines temporal evolution, dynamic profiles, and centroid drift as a separate displacement indicator; Chapter 6 studies morphological similarity, taxonomy, convergence, and divergence. Chapters 7 and 8 discuss the contribution, limits, and final findings.

2. BACKGROUND AND RELATED WORK

2.1. Science Mapping and Scientific Document Representations

Science mapping turns relations among papers, journals, authors, subject categories, or topics into spatial form. Classical bibliometric maps use citation, co-citation, bibliographic coupling, co-authorship, co-word occurrence, or journal/category assignment to make scientific systems navigable (Börner et al., 2003). Tools such as VOSviewer arrange objects so that spatial proximity reflects a chosen similarity relation (Van Eck & Waltman, 2010), while global overlay maps show how activity occupies predefined disciplinary regions (Leydesdorff et al., 2013). The important point is that every map is conditional on the relation used to build it.

Bibliometric relations encode scholarly behavior, but they are not identical to semantic proximity. Two papers may be close in citation space because they share references or communities, and two semantically similar papers may not cite one another. Text-based representations offer a complementary route. The vector-space tradition in information retrieval made documents comparable in a common numerical space (Salton et al., 1983); later dense-vector models encoded semantic associations in continuous coordinates (Mikolov et al., 2013). For this thesis, the operational value is that every retained document can be assigned coordinates, so distances, neighborhoods, centroids, dispersion, and movement can be measured.

Scientific text is not generic text. Titles and abstracts contain specialized terminology, compressed research claims, and field-specific conventions. Domain-adapted models such as SciBERT were developed because scientific corpora differ systematically from general language corpora (Beltagy et al., 2019). SPECTER introduced document-level scientific embeddings learned with citation-informed supervision (Cohan et al., 2020), and SPECTER2 extends this line within SciRepEval for scientific document representation (Singh et al., 2023). SPECTER2 is therefore a suitable representation for comparing scientific works as documents, while remaining one model-dependent representation rather than a neutral map of science.

2.2. From Semantic Location to Field Morphology

Most uses of document embeddings begin with location: where is a paper, topic, or field in semantic space? Location is important, but incomplete. A centroid summarizes the average position of a field while discarding spread, local packing, hub concentration, and spectral structure. The same field can be close to another in semantic location, far from it in morphology, and still share a domain label.

The morphological intuition has roots in work on diversity, cognitive distance, and novelty. Stirling's framework separates variety, balance, and disparity (Stirling, 2007); cognitive-distance research treats proximity and distance as conditions for learning and recombination (Nooteboom et al., 2007); studies of atypical combinations show that scientific novelty can arise from unusual bridges between existing knowledge regions (Uzzi et al., 2013). This thesis adapts that intuition to document embeddings: a field is not only a point or a label, but a distribution whose shape can be summarized.

Embedding-space morphology is the operational form of that idea. Dispersion describes how far papers spread around the field centroid. Local density describes how tightly papers pack into nearest-neighbor neighborhoods. Hubness describes whether a small subset of papers becomes disproportionately central in nearest-neighbor relations, a known issue in high-dimensional spaces (Aggarwal et al., 2001; Radovanovic et al., 2010). Spectral structure summarizes how many principal directions are needed to describe variation. Temporal evolution then asks how those same structural dimensions change across publication windows.

The gap addressed here is therefore narrower than a general critique of science maps. Existing maps answer useful questions about links, topics, classifications, and visual relationships. I use OpenAlex as an open, auditable bibliographic infrastructure (Priem et al., 2022), while treating its subfields as pragmatic analytical units rather than natural boundaries. Large bibliographic databases differ by field, document type, language, and metadata completeness (Martín-Martín et al., 2021), and OpenAlex has documented strengths and limits (Culbert et al., 2025). Within those constraints, the thesis asks what becomes visible when field-level paper distributions are measured by shape rather than only by label, location, or projected map.

3. DATA, REPRESENTATION, AND MORPHOLOGY METRICS

3.1. Corpus and Analytical Units

The empirical corpus is built from OpenAlex, an open index of scholarly works, authors, venues, institutions, and research concepts (Priem et al., 2022). I use OpenAlex because the thesis needs a corpus that can be queried at scale, linked to a documented field hierarchy, filtered through explicit metadata, and reconstructed from public infrastructure. The analytical hierarchy has three levels: subfields, fields, and domains. The direct unit of measurement is the OpenAlex subfield; fields and domains are used for aggregation and interpretation.

Each work is assigned to one subfield through its primary-topic subfield. This one-work–one-subfield design keeps the analysis auditable and avoids double counting, but it also simplifies papers with multiple intellectual affiliations. The resulting corpus should be read as a controlled analytical sample, not as a census of global science.

The synchronized title–abstract corpus covers 2000–2024 and contains 2,378,036 text-eligible works across 252 subfields, 26 fields, and 4 domains. The embedding-based analysis subset contains 2,344,927 embedded works across 241 analysis subfields. The difference reflects valid embedding availability, row alignment with metadata, and minimum-observation requirements for downstream morphology metrics.

Table 3.1

Synchronized corpus summary

Measure	**Value**
Retained works in synchronized corpus	2,378,036
Retained subfields in synchronized corpus	252
Fields represented	26
Domains represented	4
Years covered	2000–2024
Planned works	2,431,303
Total shortfall	53,267
Embedded works in analysis subset	2,344,927
Analysis subfields	241
Subfield-year cells checked	6,300
Cells reaching annual target of 400 works	5,493
Cells below annual target of 400 works	807

Note. Counts use the synchronized corpus and the row-aligned embedding analysis subset.

OpenAlex subfields are granular enough to distinguish research areas, large enough to support distributional measurement, and stable enough for comparison across the 25-year period. They are still administrative and algorithmic classifications. Coverage differences, abstract availability, classification uncertainty, and language filtering affect the sample, so the corpus defines the field distributions studied here rather than the complete universe of scientific production.

3.2. Sampling and Corpus Validation

Title–abstract records are a compromise between semantic detail and corpus-scale availability. Full text would contain richer evidence, but it is not consistently available at this scale and would introduce strong availability bias across publishers, fields, and years. Bibliographic metadata alone would be too thin for document-level morphology. I therefore use title and abstract text as the input available for a large, comparable, scientific-document corpus.

Figure 3.1

OpenAlex filtering and corpus construction pipeline for the fixed corpus used in the analysis.

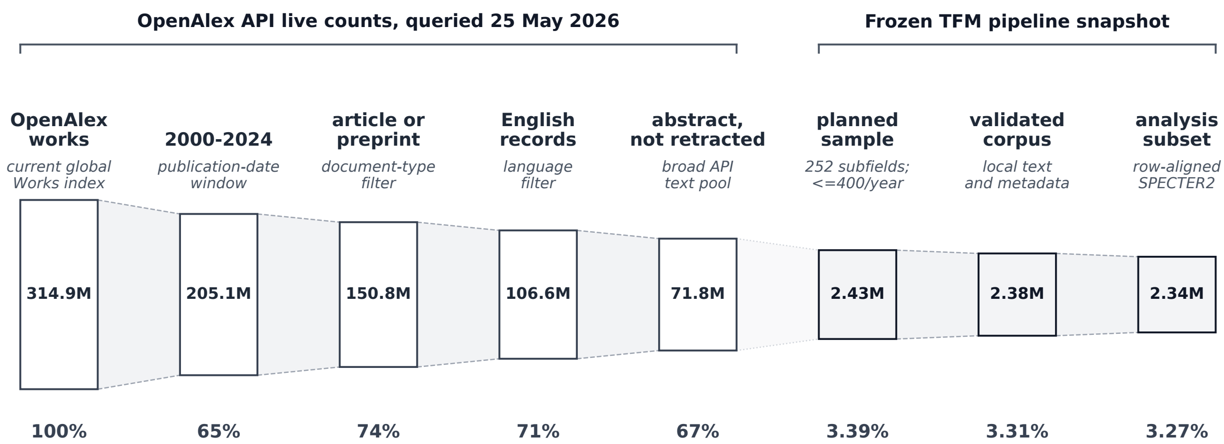


Table 3.2

Corpus eligibility filters and sampling design

Component	Active setting
Analytical unit	OpenAlex subfield
Subfield assignment	Primary-topic subfield only
Publication years	2000–2024 inclusive
Language	English
Work types	Article and preprint
Text requirement	Title and abstract required
Minimum title length	At least 5 tokens
Minimum abstract length	At least 80 tokens
Exclusions	Retracted and paratextual records
Annual sampling target	Up to 400 works per subfield-year
Full-period cap	10,000 works per subfield
Sampling controls	Oversample factor 1.75; maximum 4 backfill rounds

Note. Settings are the active corpus construction settings.

The corpus is balanced because the thesis measures shape rather than publication volume. For subfield s and year t, the planned annual sample size is

$$n_{s,t}^{planned} = \min(N_{s,t}^{eligible}, 400),$$

where $N_{s,t}^{eligible}$ is the number of works satisfying the eligibility filters in that subfield-year cell. Capacity from sparse cells is not redistributed to denser cells, so temporal structure is preserved and high-output fields do not dominate the shape estimates.

Validation checks passed for the required metadata and text conditions: no duplicate work identifiers, no out-of-period rows, no missing taxonomy identifiers, no missing title or abstract fields, no records below the minimum text thresholds, no non-English rows, and no disallowed work types. Of 6,300 checked subfield-year cells, 5,493 reach the annual target and 807 remain below target; these sparse cells are coverage limits rather than validation failures.

Table 3.3

Retained works by OpenAlex domain

Domain	Fields	Subfields	Works	Share
Life Sciences	5	42	397,808	16.73%
Social Sciences	6	58	542,774	22.82%
Physical Sciences	10	89	844,382	35.51%
Health Sciences	5	63	593,072	24.94%
Total	**26**	**252**	**2,378,036**	**100.00%**

Note. Domain counts are aggregated from retained corpus records.

The domain distribution reflects taxonomy, the annual cap, and title–abstract availability. Physical Sciences contribute the largest number of retained works, followed by Health Sciences, Social Sciences, and Life Sciences. These shares should not be read as the true distribution of global scientific output, because the sampling design deliberately reshapes observed volume to support comparable morphology measurement.

The balancing design has a cost: the analysis does not represent scientific production volume. A small, consistently sampled subfield and a large, high-output subfield can receive comparable analytical weight. Sparse cells are not filled by borrowing from later years or neighboring subfields, because imputation would add artificial temporal structure; windows that cannot support reliable metrics are excluded from complete-trajectory calculations.

3.3. Scientific Document Representation

Each retained work is represented by the concatenation of its title and abstract. The active representation model is SPECTER2, a scientific document embedding model evaluated in the SciRepEval framework (Singh et al., 2023). I use SPECTER2 because the object of analysis is scientific documents, not generic text, citation-only topology, or topic proportions.

The representation step maps each title–abstract record to a dense vector,

$$f_\theta(x_i) = z_i \in \mathbb{R}^{768},$$

where x_i is the title–abstract text of work i, f_θ is the trained embedding model, and z_i is the resulting vector. The analysis matrix is

$$Z = (z_1, \dots, z_n)^\top \in \mathbb{R}^{n\times 768}, \qquad n = 2{,}344{,}927.$$

Each row remains aligned with the same work's OpenAlex metadata: work identifier, publication year, subfield, field, and domain. This row alignment is the contract between corpus construction and morphology measurement.

All quantitative geometry is computed before dimensionality reduction. Metrics use the original L2-normalized SPECTER2 vectors and cosine distances; PCA and UMAP are used only for display. This boundary matters because projected maps can be visually useful while altering distances, densities, and neighborhood relations. The representation remains model-dependent and limited by title–abstract coverage, but it supplies a fixed measurement space for relative comparison inside the thesis.

3.4. Structural Morphology Metrics

For a subfield s, let $Z_s = \{z_i : i \in s\}$ be the set of normalized document embeddings and $c_s = n_s^{-1} \sum_{i\in s} z_i$ its centroid. The active structural morphology profile contains eight metrics in three families: global dispersion, local density and hubness, and spectral structure. Appendix A records the formulas.

Table 3.4

Structural embedding-space morphology metrics

Metric	Family	Definition	Interpretation
Centroid median	Global dispersion	Median document distance to the subfield centroid.	Typical spread around the semantic center.
Centroid IQR	Global dispersion	Interquartile range of document distances to the centroid.	Inequality of radial spread inside the subfield.
Centroid P90	Global dispersion	90th percentile of document distances to the centroid.	Outer-tail dispersion of the subfield cloud.
kNN median	Local density	Median distance from each document to its local nearest-neighbor set.	Typical local spacing among neighboring papers.
kNN CV	Local density	Coefficient of variation of local kNN distances.	Unevenness of local density.
Hub Gini	Hubness	Gini coefficient of kNN indegree counts.	Concentration of local-neighborhood centrality.
PCA D80	Spectral structure	Number of principal components needed to explain 80% variance.	Effective dimensionality of the subfield cloud.
PCA entropy	Spectral structure	Entropy of the normalized PCA spectrum.	Evenness of variance across latent directions.

Note. These eight metrics define the active structural morphology profile. Net semantic displacement is measured separately with early–late centroid drift and is not included in the profile matrix.

Dispersion metrics describe the distribution of cosine distances from papers to the subfield centroid: typical spread, middle-spread heterogeneity, and the outer semantic periphery. A high median centroid distance means that the typical retained paper sits farther from the subfield center. A high centroid-distance IQR means that the middle of the field is uneven. A high 90th percentile means that the semantic periphery extends farther, even if the center remains stable.

Local-density metrics use within-subfield k-nearest-neighbor distances (Cover & Hart, 1967), with $k = 15$ unless fewer non-self neighbors are available. Median kNN distance describes local packing rather than field radius: two subfields can have similar centroid dispersion while one has much closer local neighborhoods. The kNN distance coefficient of variation describes unevenness in local packing. Hubness is measured by the Gini coefficient of directed kNN in-degree counts, so high hubness means that a smaller set of papers appears disproportionately often as neighbors of other papers.

Spectral structure is measured through PCA on normalized embeddings (Jolliffe & Cadima, 2016): D_{80} counts the components required to explain 80 percent of variance, and spectral entropy (Shannon, 1948) measures how evenly variance is distributed across positive-variance directions. These are operational summaries of the variance spectrum. They should not be read as estimates of a field's true dimensionality, but they help distinguish fields whose variation is concentrated in fewer directions from fields whose variation is spread more broadly.

Net semantic displacement is measured separately as early–late centroid drift, the cosine distance between the 2000–2004 and 2020–2024 subfield centroids. It is not part of the structural profile matrix used for static profiles, PCA, clustering, or morphological distance. Its role is complementary: it reports movement of the subfield center after the structural morphology analysis has been defined.

3.5. Analytical Workflow

The workflow has four stages. First, subfield-level structural morphology profiles are computed from row-aligned SPECTER2 embeddings and then aggregated to fields and domains by averaging constituent subfield profiles. Second, static comparisons use robust standardization to compare profiles across domains, fields, and subfields. Third, temporal analyses recompute the same eight structural metrics within five windows: 2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024. Fourth, relational analyses compute Euclidean distances between robust-scaled field profiles to study morphological similarity, nearest neighbors, convergence, and divergence.

Robust standardization is used because the metric profiles contain outliers by design: unusually broad, compact, sparse, dense, hub-concentrated, or spectrally complex subfields are part of the signal. For metric m, subfield values are centered by the cross-subfield median and scaled by the interquartile range before aggregation or distance calculation, following the same robust-scale logic used in resistant scale estimation (Rousseeuw & Croux, 1993). This keeps extreme profiles visible without allowing one metric scale to dominate the profile distance.

Aggregation is descriptive. A field profile is the mean of its constituent subfield profiles after standardization; a domain profile is the corresponding mean across subfields within that domain. These averages support broad comparison, but they are not treated as homogeneous entities. The static chapter therefore moves from domains to fields and subfields, while the temporal chapter checks how the same structural dimensions evolve.

The reproducible pipeline follows the same sequence as the analysis: taxonomy retrieval, annual count estimation, sample planning, work retrieval, corpus validation, embedding validation, row-aligned matrix construction, metric computation, temporal window computation, profile aggregation, and figure/table generation. Central numerical claims come from stored pipeline outputs, not manual case selection. Clustering is run on profile representations rather than projection coordinates.

These operations define the evidential boundary of the thesis. The metrics describe structure under a specific corpus, taxonomy, representation model, and set of geometric summaries. They do not measure publication volume, impact, quality, institutional importance, or causal mechanisms. Their purpose is narrower: to make the shape, change, and relational profiles of scientific fields comparable.

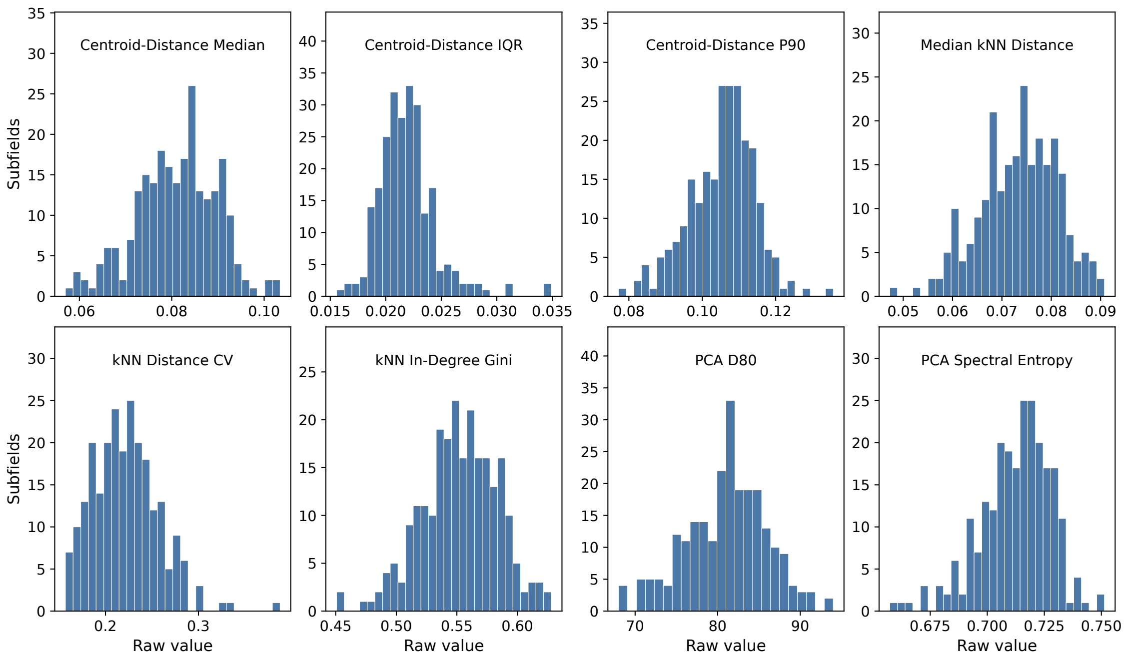
4. STATIC MORPHOLOGY ACROSS SCIENTIFIC FIELDS

4.1. Exploratory Metric Diagnostics

Before comparing fields, the eight raw structural metrics are inspected directly. The distributions are not Gaussian: centroid-distance IQR is right-skewed, kNN distance CV has a visible upper tail, and the PCA metrics are bounded or partly discrete. Other metrics are more symmetric but still contain meaningful extremes. This distributional shape justifies robust median/IQR scaling in the static profile analysis, because the most unusual subfields are substantive cases rather than errors to be smoothed away.

Figure 4.1

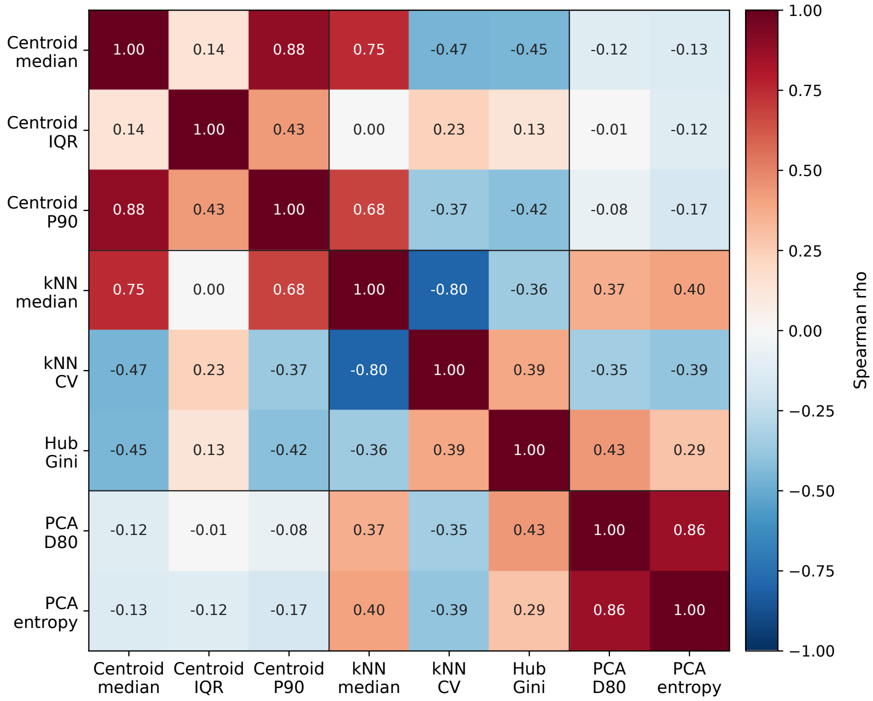
Raw subfield distributions of the eight structural morphology metrics. Dashed vertical lines mark medians.



The correlation structure shows related but not redundant metric families. Centroid median and centroid P90 are strongly correlated ($\rho = 0.883$), as expected for two measures of radial spread. PCA D_{80} and PCA entropy are also strongly related ($\rho = 0.856$), because both summarize the variance spectrum. The largest negative relation is between kNN median distance and kNN CV ($\rho = -0.798$): fields with closer local neighborhoods tend to have more uneven local packing. Cross-family correlations are weaker, supporting the decision to keep dispersion, local density, hubness, and spectral structure as separate dimensions.

Figure 4.2

Spearman correlation matrix for the eight raw structural morphology metrics.



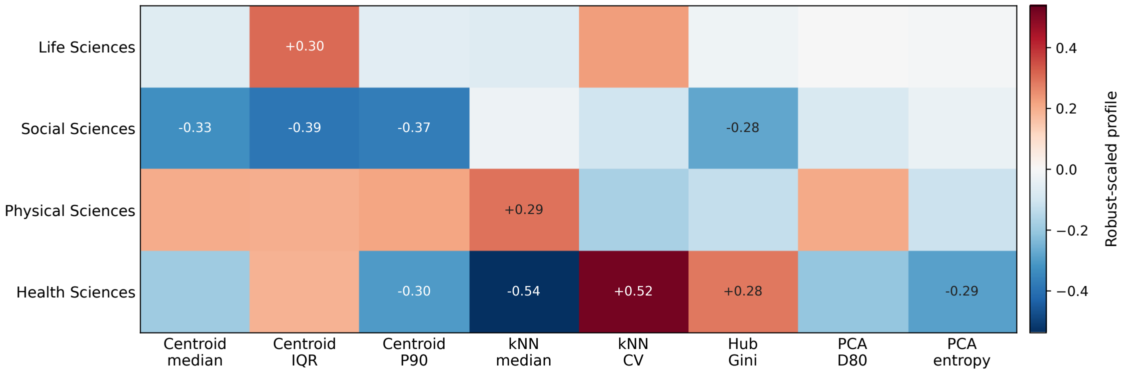
4.2. Domain Tendencies

Static comparison assigns each subfield one full-period structural morphology profile using the eight metrics defined in Chapter 3. Subfield profiles are the direct unit of measurement; field and domain profiles are averages over constituent subfields after robust standardization. For the heatmaps and boxplots below, red indicates above-median values for that metric and blue indicates below-median values.

At the broadest scale, domains show morphological tendencies. Physical Sciences are generally more dispersed and spectrally extended. Health Sciences show tighter local neighborhoods and stronger local concentration. Social Sciences and Life Sciences sit closer to the center of most structural metrics, with specific departures rather than a single dominant profile.

Figure 4.3

Domain-level standardized structural morphology profiles. Values are visual contrasts across the four domain profiles, not inferential statistics.



The domain view is useful because it gives the first answer to the static part of the second research question: broad areas of science differ in shape. It is also limited, because four averages can easily hide the heterogeneity that matters for the thesis.

4.3. Field and Subfield Heterogeneity

Field profiles break the clean domain story. Physical Sciences include dispersed fields such as Physics and Astronomy, Computer Science, Engineering, and Environmental Science, but also more compact fields such as Energy and Mathematics. Health Sciences include dense fields such as Dentistry and Nursing, while Medicine sits closer to the center of the field distribution. Domains therefore have tendencies, not uniform morphologies.

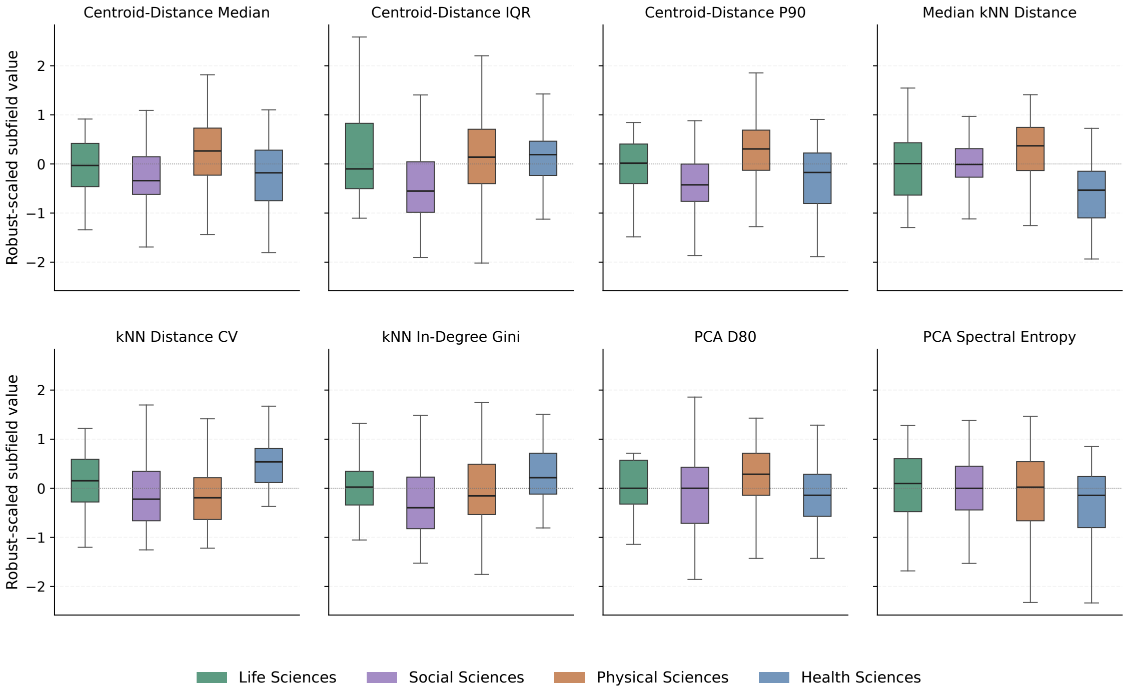
Figure 4.4

Field-level standardized structural morphology profiles. Rows are fields ordered by parent OpenAlex domain.

The most visible contrast is local structure. Chemistry, Physics and Astronomy, Materials Science, Engineering, and Environmental Science have high median kNN distances, meaning that even near neighbors tend to be farther away. Dentistry, Nursing, Immunology and Microbiology, Veterinary, and Health Professions have lower median kNN distances. This does not mean that one group is more important or more coherent in a substantive sense; it means that their retained title–abstract records are packed differently in the SPECTER2 space.

Figure 4.5

Subfield-level standardized metric distributions by domain for the eight structural metrics.



The boxplots show why morphology cannot be inferred from a domain label. Physical Sciences have the highest median subfield dispersion and kNN distance, but some physical subfields are compact. Health Sciences have lower local distances on average, but hubness varies widely. Social Sciences contain both compact applied areas and highly atypical humanities subfields. The field and subfield levels are therefore not details after the domain result; they are the evidence that taxonomy and morphology only partly align.

4.4. Metric Extremes and Profile Space

Selected extremes make the same point with auditable cases. Nuclear and High Energy Physics and Artificial Intelligence sit at the high end of median centroid distance, while Applied Microbiology and Biotechnology and General Energy sit at the compact end. Molecular Medicine and Philosophy have unusually high centroid-distance IQR; Research and Theory and Algebra and Number Theory sit at the opposite end. Local density, hubness, and spectral structure cut across those patterns, so a field can be broad without being hub-dominated, and a compact field can still be locally uneven.

Table 4.1

Selected static structural-metric extremes at subfield level

Metric	High end	Low end
Centroid median	Nuclear and High Energy Physics (+1.82); Artificial Intelligence (+1.78)	Applied Microbiology and Biotechnology (-2.08); General Energy (-1.87)
Centroid IQR	Molecular Medicine (+5.02); Philosophy (+5.02)	Research and Theory (-2.34); Algebra and Number Theory (-2.02)
Centroid P90	Philosophy (+2.50); Nuclear and High Energy Physics (+1.85)	Research and Theory (-2.51); General Energy (-2.06)
kNN median	Molecular Biology (+1.55); Materials Chemistry (+1.41)	Applied Microbiology and Biotechnology (-2.44); Internal Medicine (-1.94)
kNN CV	Applied Microbiology and Biotechnology (+3.49); Toxicology (+2.43)	History (-1.25); Anthropology (-1.25)
Hub Gini	General Materials Science (+1.75); Electrochemistry (+1.62)	Artificial Intelligence (-2.44); Computer Vision and Pattern Recognition (-2.35)
PCA D80	Classics (+1.86); Religious studies (+1.71)	Business and International Management (-1.86); Computational Mathematics (-1.86)
PCA entropy	Classics (+1.94); Religious studies (+1.81)	Human Factors and Ergonomics (-3.05); Instrumentation (-2.70)

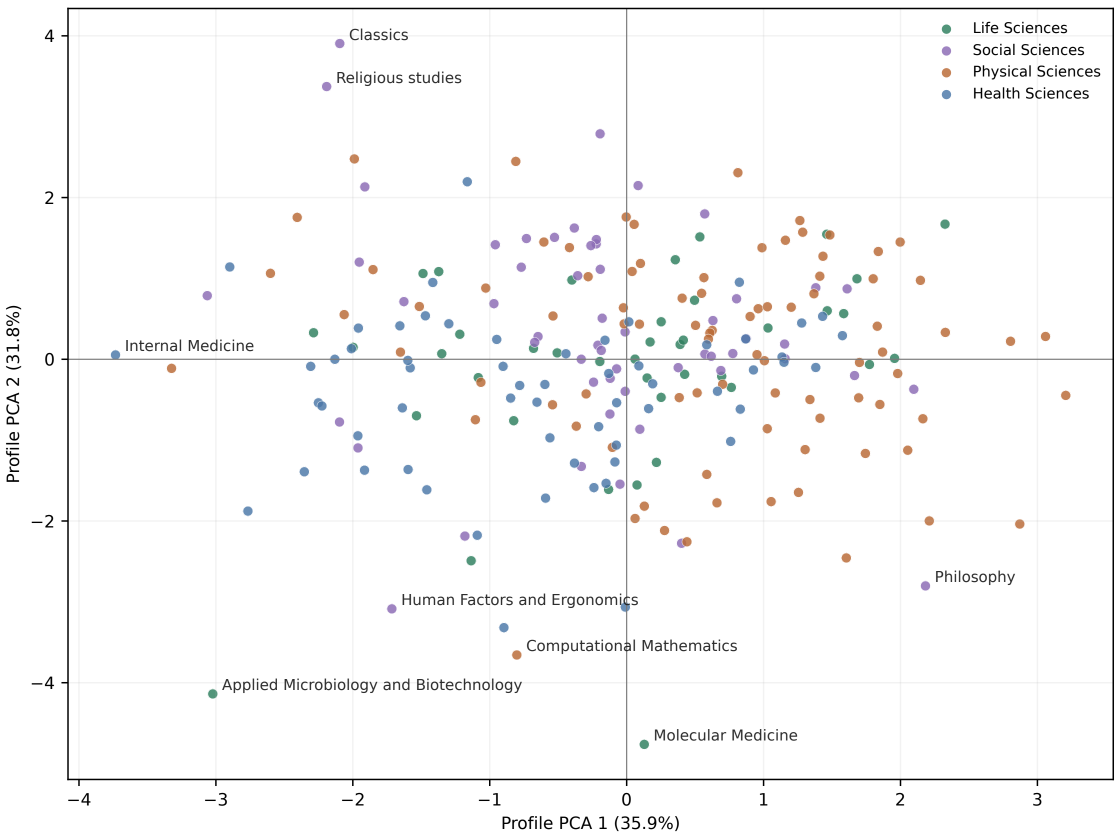
Note. Values in parentheses are robust-scaled subfield profile values under the active eight-metric structural specification.

The extremes also show why a single composite score would weaken the argument. Human Factors and Ergonomics is unusual in spectral entropy, Applied Microbiology and Biotechnology in local-density variability, Classics and Religious studies in spectral extension, and Artificial Intelligence in low hub concentration. Keeping the metric families separate preserves those distinctions instead of collapsing them into one broad atypicality ranking.

The exploratory PCA map in Figure 4.6 summarizes standardized structural profiles, not paper-level semantic locations. The first two axes explain 35.9 percent and 31.8 percent of the standardized profile variance. Domains overlap in the center, while specific subfields form the most informative extremes.

Figure 4.6

Exploratory PCA map of subfield structural morphology profiles. Points are subfields represented by the eight standardized structural metrics.



The static result is deliberately modest: domains help orient the comparison, but they do not determine field shape. Broad domain tendencies coexist with large within-domain variation and cross-domain overlap in metric-profile space; Appendix B retains selected UMAP maps only as visual diagnostics. This establishes static morphology as measurable but not neatly taxonomic, which motivates the next chapter's shift from full-period shape to structural change over time.

5. TEMPORAL EVOLUTION AND DYNAMIC PROFILES

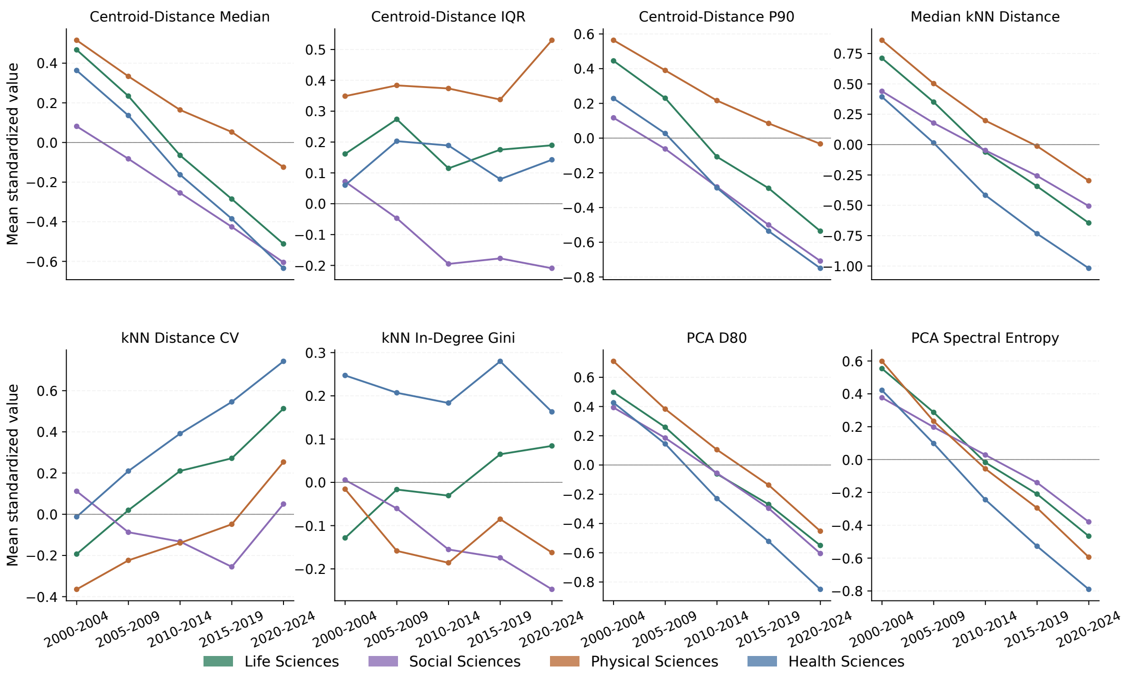
5.1. Temporal Trajectories of Structural Metrics

Temporal morphology is measured through five fixed windows: 2000–2004, 2005–2009, 2010–2014, 2015–2019, and 2020–2024. Within each window, I recompute the same eight structural metrics used in the static profile. Windowed plots use robust-scaled values computed across all completed subfield-window observations. The temporal stage is nearly complete: 1,204 of 1,205 expected subfield-window rows were computed; Computational Mathematics in 2000–2004 is excluded where first–last comparison requires a complete trajectory.

The domain view gives the broad orientation. Across all subfields, mean median centroid distance declines from 0.0858 in 2000–2004 to 0.0750 in 2020–2024, mean median kNN distance declines from 0.0923 to 0.0756, and PCA D_{80} falls from 81.3 to 71.1. At the same time, kNN distance variability rises. The population-level pattern is therefore closer local packing with more uneven local distance structure, not a simple contraction.

Figure 5.1

Domain-level temporal trajectories across five-year windows. Values are mean standardized subfield-window structural metrics within each domain.



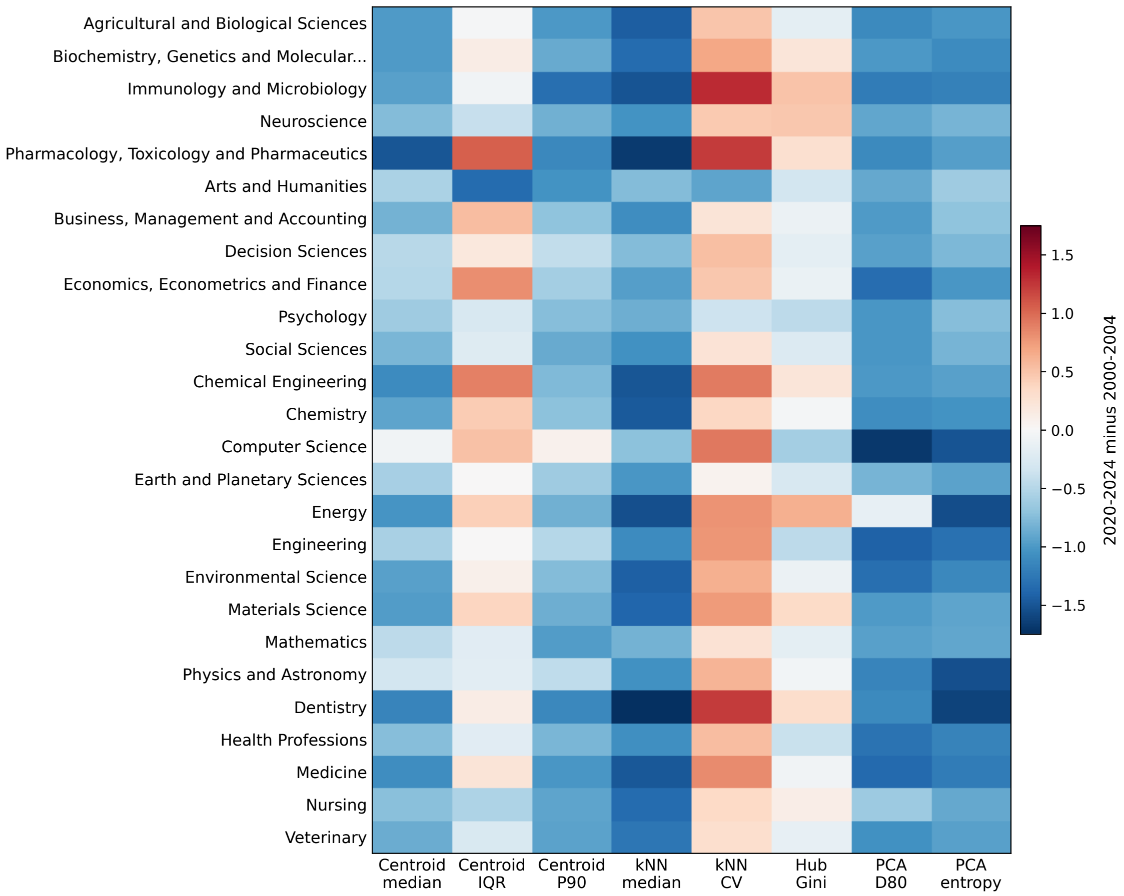
Physical Sciences remain the most dispersed domain for most of the period and retain the highest average spectral dimensionality. Health Sciences move furthest downward in local distance and PCA D_{80}, ending with the tightest local neighborhoods and the lowest average dimensionality. Life Sciences follow the same direction with a clearer rise in kNN distance variability. Social Sciences are less extreme in domain averages, which again makes the subfield and field levels necessary.

5.2. Fields and Subfields in Motion

Field-level first–last deltas show where the average trajectory is strongest. Pharmacology, Toxicology and Pharmaceutics, Dentistry, Chemical Engineering, Medicine, and Energy have especially strong declines in median centroid distance. Dentistry, Energy, Immunology and Microbiology, Medicine, and Chemistry show large declines in median kNN distance. Hubness does not follow the same direction: Energy and Immunology and Microbiology move upward in hub concentration, while Computer Science, Engineering, Psychology, and Health Professions move downward.

Figure 5.2

Field-level temporal change from 2000–2004 to 2020–2024. Cells are final-minus-initial standardized structural values.

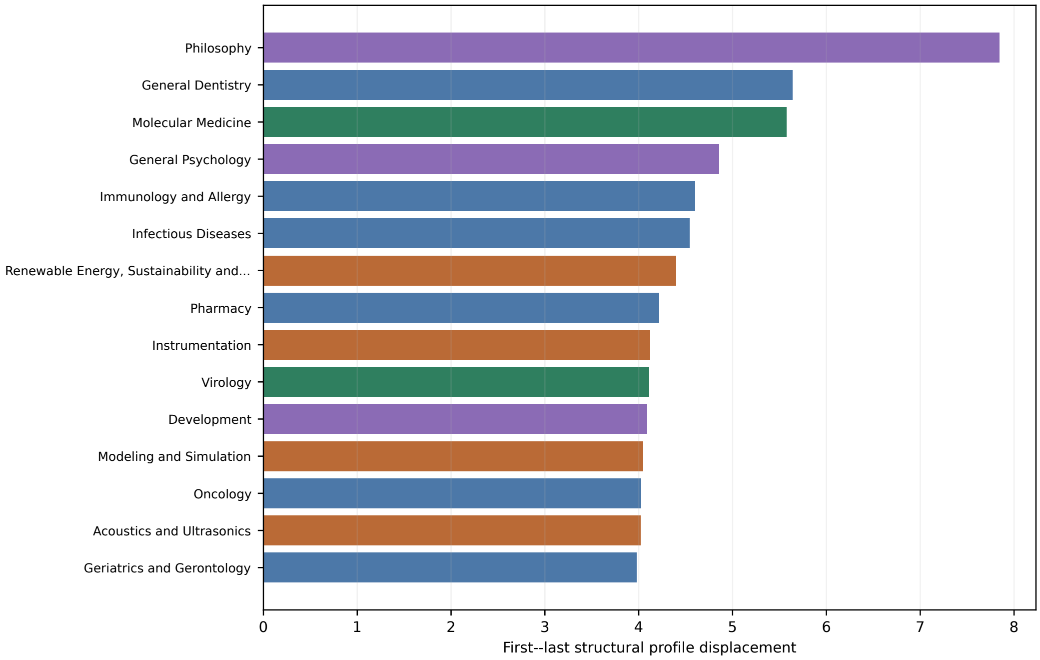


The strongest first–last profile changes are concentrated in a small set of subfields. Philosophy has the largest windowed structural displacement, followed by General Dentistry, Molecular Medicine, General Psychology, Immunology and Allergy, Infectious

Diseases, Renewable Energy, Sustainability and the Environment, Pharmacy, Instrumentation, and Virology. The ranking measures Euclidean distance between initial and final standardized profiles across the eight windowed structural metrics.

Figure 5.3

Most dynamic subfields across windowed structural metrics. The ranking measures total first–last profile displacement.



The drivers behind these cases are mixed. Philosophy is dominated by a sharp decrease in centroid-distance IQR, while Molecular Medicine moves in the opposite direction on that same metric. General Dentistry is driven by a decline in PCA spectral entropy. General Psychology changes primarily through reduced kNN distance variability, whereas Immunology and Allergy and Virology change through increases in that metric. Temporal movement is therefore structured, but not one common pattern.

5.3. Net Semantic Displacement

The eight-metric structural profile describes shape. A separate question is whether the semantic center of a subfield moves across the period. I measure this complementary quantity with early–late centroid drift, the cosine distance between each subfield's 2000–2004 and 2020–2024 centroids. Across analysis subfields, the median drift is 0.0045 and the maximum is 0.0384.

Table 5.1

Highest and lowest early–late centroid drift by subfield

Group	Subfield	Field	Domain	Drift
Highest	Human Factors and Ergonomics	Social Sciences	Social Sciences	0.038
Highest	Computational Mathematics	Mathematics	Physical Sciences	0.019
Highest	Modeling and Simulation	Mathematics	Physical Sciences	0.016
Highest	Ecological Modeling	Environmental Science	Physical Sciences	0.016
Highest	Acoustics and Ultrasonics	Physics and Astronomy	Physical Sciences	0.015
Highest	Computer Vision and Pattern Recognition	Computer Science	Physical Sciences	0.014
Highest	Energy Engineering and Power Technology	Energy	Physical Sciences	0.013
Highest	Health Information Management	Health Professions	Health Sciences	0.013
Highest	Signal Processing	Computer Science	Physical Sciences	0.013
Highest	Process Chemistry and Technology	Chemical Engineering	Physical Sciences	0.012
Lowest	Mathematical Physics	Mathematics	Physical Sciences	0.001
Lowest	Algebra and Number Theory	Mathematics	Physical Sciences	0.001
Lowest	Geometry and Topology	Mathematics	Physical Sciences	0.002
Lowest	Paleontology	Earth and Planetary Sciences	Physical Sciences	0.002
Lowest	Anthropology	Social Sciences	Social Sciences	0.002
Lowest	Ceramics and Composites	Materials Science	Physical Sciences	0.002
Lowest	Mechanics of Materials	Engineering	Physical Sciences	0.002
Lowest	Numerical Analysis	Mathematics	Physical Sciences	0.002
Lowest	Mechanical Engineering	Engineering	Physical Sciences	0.002
Lowest	Oral Surgery	Dentistry	Health Sciences	0.002

Note. Centroid drift measures net semantic displacement between the early and late parts of a subfield's publication history. Low values indicate the most stable centroids in this corpus. Drift is reported separately and is not part of the structural morphology profile.

The largest net displacement belongs to Human Factors and Ergonomics, followed by Computational Mathematics, Modeling and Simulation, Ecological Modeling, Acoustics and Ultrasonics, and Computer Vision and Pattern Recognition. The most stable centroids include Mathematical Physics, Algebra and Number Theory, Geometry and Topology, Paleontology, and Anthropology. These cases are not added back into the structural profile. They answer a different diagnostic question: how far the center of the subfield moved, after structural shape and structural evolution have been measured separately.

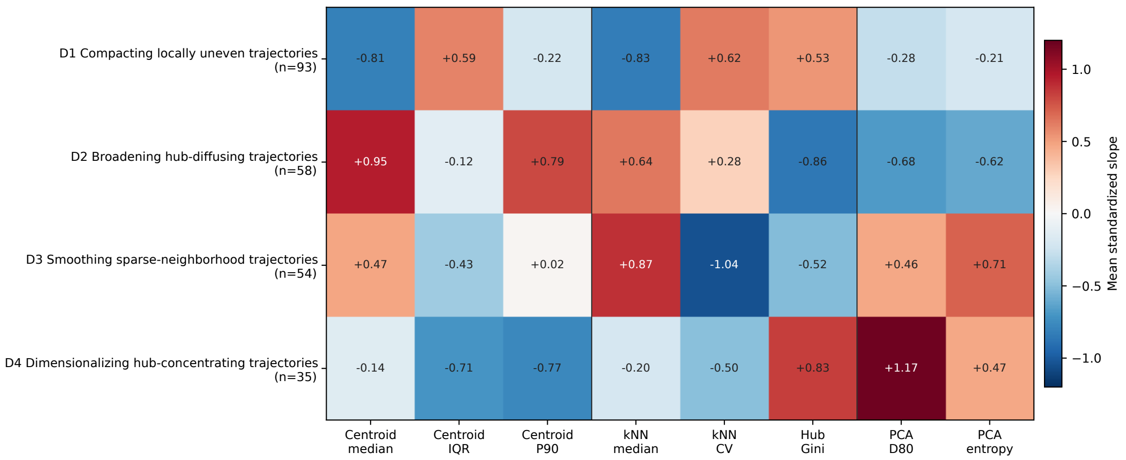
5.4. Static Shape Versus Temporal Trajectories

The useful part of clustering is compression. Static morphology clusters the 241 analysis subfields by eight robust-scaled full-period structural metrics. Dynamic morphology clusters the 240 complete subfields by the slopes of the same eight metrics across windows. In both cases, clusters are interpretive summaries rather than natural kinds or a replacement for continuous profile space.

The selected static solution uses Ward hierarchical clustering, Euclidean distance, robust median/IQR scaling, and $k = 5$ (Ward, 1963). Its silhouette is 0.174 (Rousseeuw, 1987), so static morphology is better read as a continuous profile space than as a set of separated classes. The selected dynamic solution uses average-link hierarchical clustering with correlation distance at $k = 4$. It has silhouette = 0.396 and mean subsample ARI = 0.725 (Hubert & Arabie, 1985), making it a clearer and more stable descriptive compression.

Figure 5.4

Dynamic temporal-trajectory typology profiles. Values are mean standardized linear slopes of the eight windowed structural metrics.



The four dynamic groups describe movement patterns. D1, with 93 subfields, contracts in centroid and kNN distance while local unevenness and hub concentration rise. D2, with 58 subfields, broadens while hubness and spectral complexity decline. D3, with 54 subfields, becomes sparser in local distance while smoothing local unevenness and hubness. D4, with 35 subfields, increases PCA dimensionality and hubness while outer spread and local variability fall. These labels are shorthand for slope profiles, not a new taxonomy.

The dynamic typology should still be read as a vocabulary of trajectories, not as a new classification of science. The groups depend on the selected corpus, representation, metrics, windows, scaling, and clustering procedure; their value is descriptive compression, not causal explanation. The main temporal result is that structural profiles move in heterogeneous but measurable ways, while centroid drift supplies a separate view of semantic displacement.

6. MORPHOLOGICAL SIMILARITY AND TAXONOMY

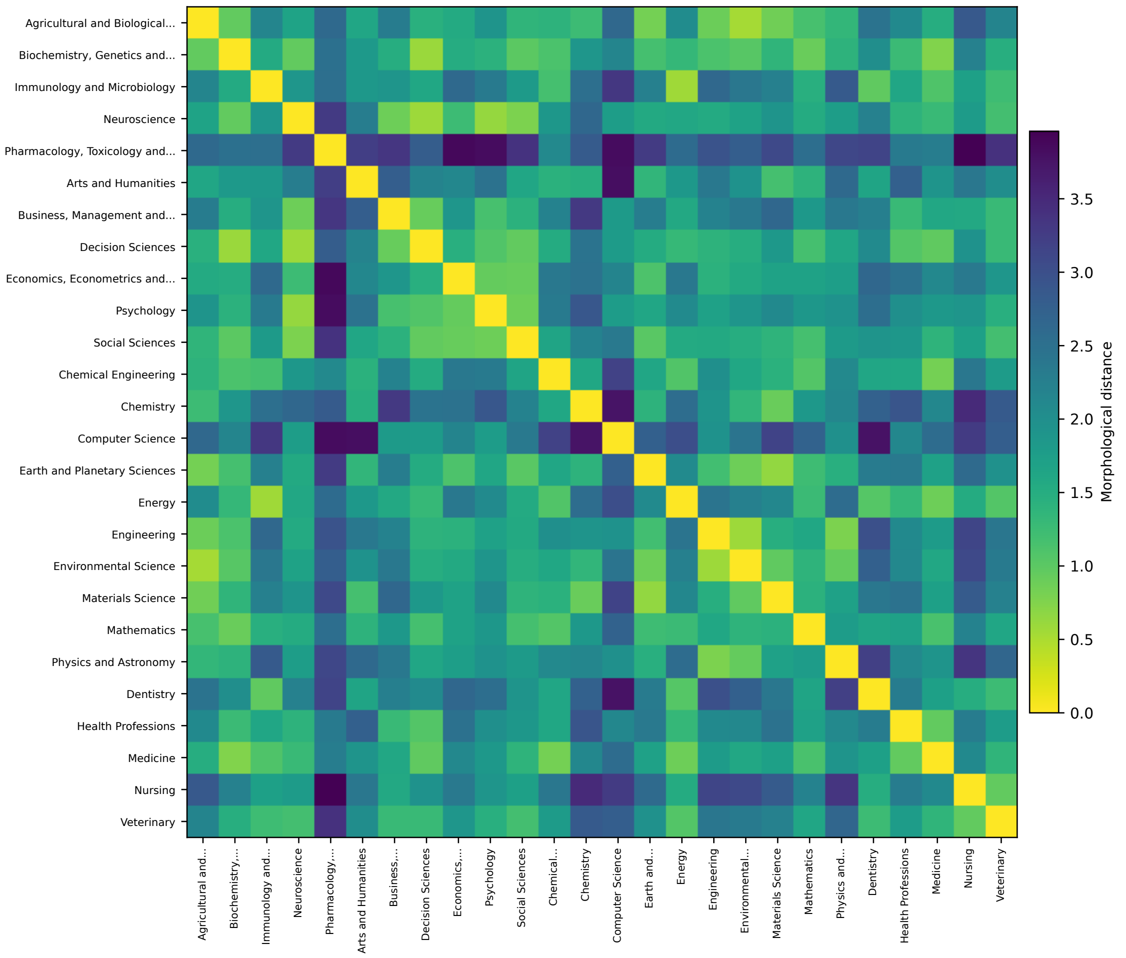
6.1. Static Similarity Across Fields

Two fields are morphologically similar when their robust-scaled structural metric profiles are close. This is not document-level semantic similarity: two fields can have similar structural profiles without using the same vocabulary, citing the same work, or studying the same topics. Static similarity uses the eight-metric structural profile; field and domain profiles are averages of robust-scaled subfield profiles; distances are Euclidean distances between those profiles.

The static field-level distance matrix tests how far morphology follows OpenAlex domains. If domains fully organized morphology, the diagonal blocks would be uniformly light and off-block regions uniformly dark. The observed pattern is weaker: some domain blocks are internally coherent, but many local profile-space relations cross domain boundaries.

Figure 6.1

Pairwise morphological distance between fields using the eight structural metrics. Lower values indicate more similar robust-scaled morphology profiles.



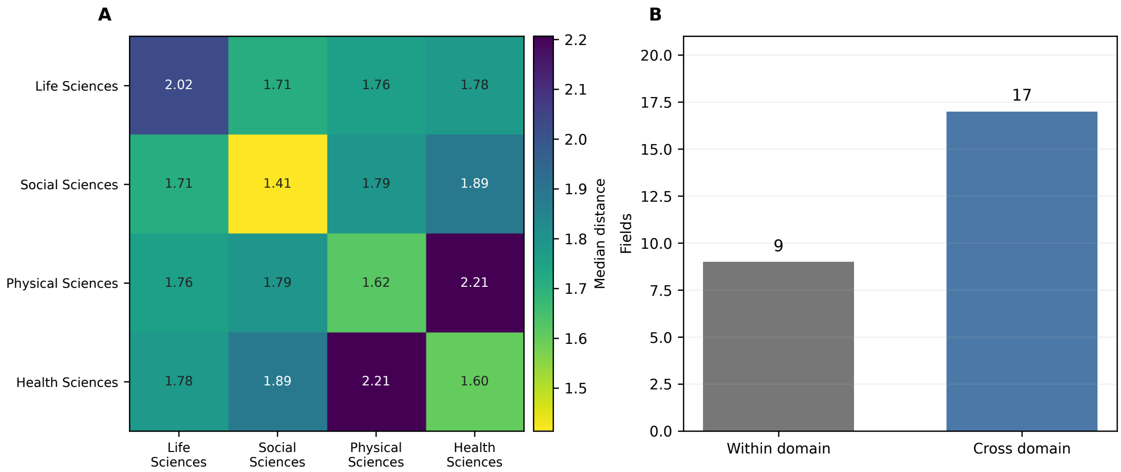
The closest Euclidean field pairs are not simply within-domain neighbors. Agricultural and Biological Sciences is closest to Environmental Science (0.530); Energy is close to Immunology and Microbiology (0.562); Decision Sciences is close to Neuroscience (0.575); Engineering is close to Environmental Science (0.587); and Biochemistry, Genetics and Molecular Biology is close to Decision Sciences (0.594). These links do not imply shared content. They mean that the fields have similar standardized levels of compactness, dispersion, hubness, and spectral structure.

6.2. Taxonomy, Bridges, and Isolates

The domain-pair summary separates within-domain and cross-domain structure. Within-domain field pairs have a lower median distance than cross-domain pairs, 1.62 versus 1.86, but the relationship is uneven. Social Sciences are the most internally coherent domain, and the closest cross-domain medians involve Life Sciences with Social Sciences and Life Sciences with Physical Sciences. At field level, 17 of 26 fields have their nearest morphological neighbor outside their own domain.

Figure 6.2

Domain-pair morphological distance structure. Panel A reports median field-pair distances by OpenAlex domain pair; Panel B counts within-domain and cross-domain nearest neighbors.



Several bridges are substantively interesting in profile space: Agricultural and Biological Sciences with Environmental Science; Energy with Immunology and Microbiology; Decision Sciences with Neuroscience; and Medicine with Biochemistry, Genetics and Molecular Biology. These are morphological bridges only. They show similar profile shape, not collaboration, topical merger, or citation linkage. The isolates are equally important: Pharmacology, Toxicology and Pharmaceutics has the largest nearest-neighbor distance, with Chemical Engineering still 2.10 away; Computer Science follows, with Neuroscience 1.75 away.

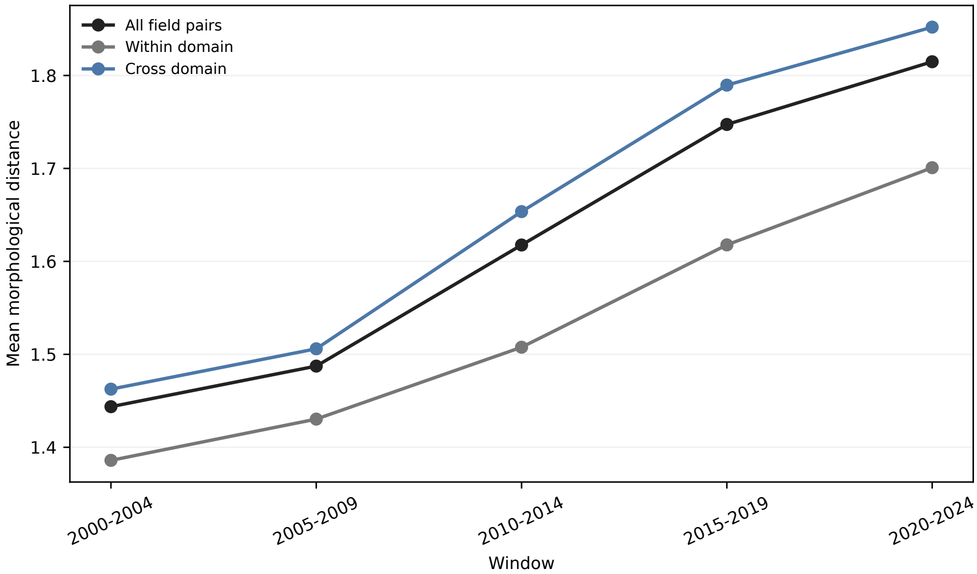
6.3. Convergence and Divergence

Temporal convergence and divergence use the eight structural metrics recomputed inside each five-year window. Operationally, convergence means that the distance between two windowed profiles decreases from 2000–2004 to 2020–2024; divergence means that it increases. Centroid drift is not included in these pairwise distances, because it is a separate semantic-displacement indicator rather than a structural profile dimension.

The temporal picture is not global convergence. Across all 325 field pairs, mean Euclidean distance rises from 1.44 in 2000–2004 to 1.81 in 2020–2024. The same upward movement appears within domains, from 1.39 to 1.70, and across domains, from 1.46 to 1.85. Field-level morphology space becomes more separated on average, even though many individual pairs move closer.

Figure 6.3

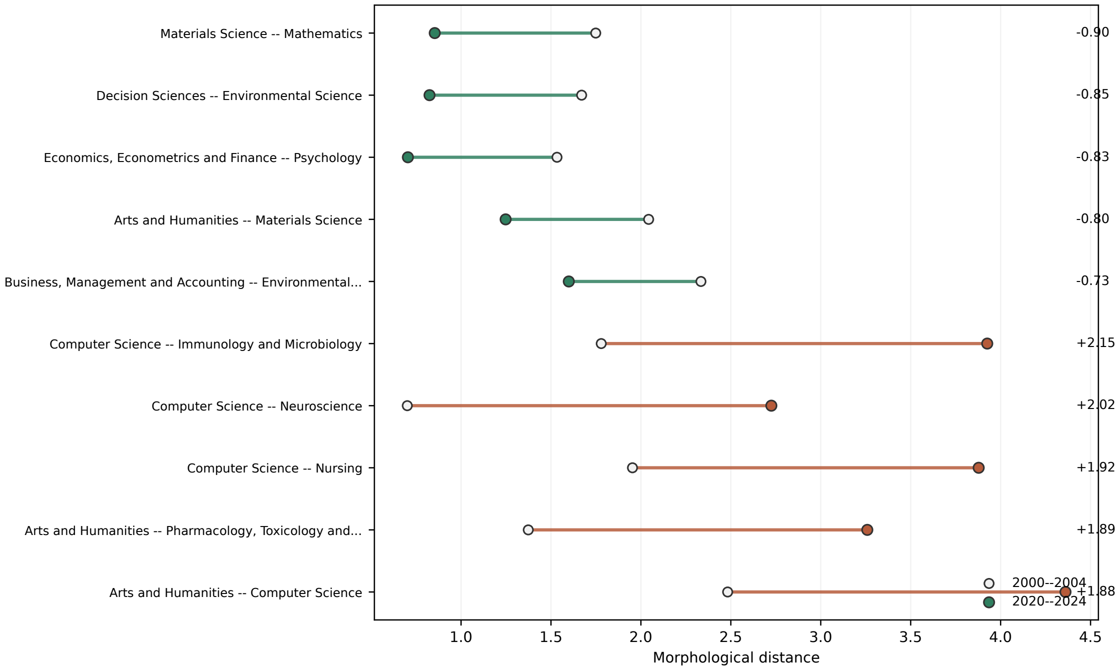
Mean field-pair morphological distance across five-year windows using the eight windowed structural metrics.



Selective convergence remains visible: 97 field pairs decrease in distance, while 228 increase. The strongest convergences tend to involve shared declines in local distance and spectral dimensionality. Materials Science and Mathematics move from 1.75 to 0.85; Decision Sciences and Environmental Science from 1.67 to 0.82; and Economics, Econometrics and Finance and Psychology from 1.53 to 0.70. The diverging side has a clearer protagonist: Computer Science appears in nine of the ten largest field-level Euclidean divergences.

Figure 6.4

Strongest field-level convergence and divergence pairs. Each line compares initial and final windowed profile distances; signed labels identify the largest decreases and increases in morphological distance.



The same distance change can come from different metric movements. Some convergences reflect both fields becoming locally denser or lower-dimensional. Some divergences occur because one field changes sharply while the other follows the population average. Computer Science is the clearest case: its large drops in PCA dimensionality and spectral entropy separate its windowed profile from many fields that also become locally denser but do not move in the same spectral direction.

The relational result is the most direct answer to the third research question. OpenAlex domains still matter, but morphology produces cross-domain affinities, isolated fields, and temporal differentiation that taxonomy alone does not show.

7. DISCUSSION

7.1. Interpretation and Contribution

The thesis shows that field morphology adds a measurement layer between taxonomy, semantic location, and visualization. A domain label tells us how a work is classified; a centroid gives an average location; a projection gives a readable display. Structural morphology asks how the retained papers are arranged around one another: how dispersed, locally dense, hub-concentrated, and spectrally extended they are.

The empirical chapters support the same interpretation from three directions. Static profiles show broad domain tendencies but large within-domain variation. Temporal profiles show structured change in the same eight structural dimensions without one universal direction. Relational profiles show that OpenAlex domains organize part of morphology but do not determine nearest neighbors, bridges, isolates, convergence, or divergence.

The contribution is therefore not another field classification. It is a way to keep labels, locations, shapes, trajectories, and relations analytically separate. That separation matters because visually close regions, taxonomically related fields, and morphologically similar fields need not be the same. Centroid drift further separates semantic displacement from structural morphology: a field can move in semantic location without having the same structural change as another field.

7.2. Limits of the Evidence

The evidence is conditional on the OpenAlex corpus, its primary-topic taxonomy, the balanced sampling design, SPECTER2, and the chosen metrics. Coverage differs across fields, years, languages, document types, and metadata practices; title–abstract embeddings compress scientific documents while missing many contextual features; and high-dimensional geometry brings anisotropy, density variation, and hubness (Aggarwal et al., 2001; Radovanovic et al., 2010).

The strongest claims are therefore comparative. I can say that one field is more dispersed than another under the fixed representation, that one pair of fields becomes closer in the selected profile distance, or that temporal slope profiles cluster more clearly than static profiles. I cannot infer why those changes occurred. Convergence and divergence are profile-distance terms, not claims about collaboration, citation flows, disciplinary merger, or epistemic separation.

7.3. Implications and Future Validation

Future work should first test model and corpus robustness. The same sampling plan, metric code, and aggregation rules can be rerun with alternative scientific embedding models, multilingual or full-text representations, and other bibliographic infrastructures. Changing one component at a time would show which findings are stable properties of the framework and which depend on representation or coverage.

External validation should compare morphology with evidence that was deliberately excluded from the metric construction: citation networks, journal structures, funding categories, author mobility, and expert assessments. Agreement would strengthen the interpretation of specific cases; disagreement would be equally useful because it would show where geometric shape and other forms of scientific organization separate.

Temporal and substantive validation should focus on the strongest structural shifts, convergences, divergences, and centroid-drift cases. Five-year windows make the comparison stable, but they can smooth short shocks and database changes. Annual, rolling, and event-centered trajectories, combined with expert inspection of extreme cases, would test whether morphology highlights meaningful field transformations or mainly representation artifacts.

8. CONCLUSION

This thesis measured scientific fields as distributions of papers in a dense scientific document embedding space. I built a balanced OpenAlex title–abstract corpus for 2000–2024, represented retained works with row-aligned SPECTER2 embeddings, and computed an eight-metric structural morphology profile in the original 768-dimensional space. The metrics summarized dispersion, local density, hubness, and spectral structure; projections were used only for display.

The results show that domains have morphological tendencies but do not determine field shape, that structural profiles change over time in heterogeneous but measurable ways, and that morphological similarity only partly follows OpenAlex taxonomy. At field level, 17 of 26 fields have their nearest morphological neighbor outside their own OpenAlex domain. Mean field-pair structural distance rises from 1.44 in 2000–2004 to 1.81 in 2020–2024, indicating average differentiation alongside selective convergence between specific field pairs.

Centroid drift is retained as a separate measure of net semantic displacement. It complements the structural analysis without entering the active morphology profile, PCA, clustering, or distance matrices. This separation is the main methodological point: morphology is the eight-dimensional structural shape, temporal evolution is the change of that shape through windows, and semantic displacement is the movement of the center.

The contribution is limited but useful. Once papers are represented in a common scientific-document space, fields can be compared as distributions rather than only as labels, centroids, or visual regions. That separation makes it possible to ask where a field is, how it is shaped, how it changes, and which other fields it resembles without treating any one description as the whole structure of science.

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A. EMBEDDING-SPACE METRIC FORMULATIONS

This appendix records the formal definitions behind the active eight-metric structural morphology profile. All quantities are computed on L2-normalized SPECTER2 document embeddings in the original embedding space. For a subfield s, let $Z_s = \{z_i : i \in s\}$, $n_s = |Z_s|$, $c_s = n_s^{-1} \sum_{i \in s} z_i$, and $r_i = d(z_i, c_s)$, where $d(\cdot, \cdot)$ is cosine distance.

Global dispersion. The three centroid-based dispersion metrics are

$$\text{median}_{i \in s}(r_i), \qquad Q_{0.75}(r) - Q_{0.25}(r), \qquad Q_{0.90}(r).$$

They correspond to typical distance from the centroid, middle-spread heterogeneity, and the outer semantic periphery.

Local density and hubness. Let $N_k(i)$ be the directed set of within-subfield nearest neighbors for paper i, with $k = 15$ unless the subfield has fewer available non-self neighbors. Let

$$D_k(s) = \{d(z_i, z_j) : i \in s,\ j \in N_k(i)\}, \qquad h_j = \sum_{i \in s} \mathbf{1}\{j \in N_k(i)\}.$$

The local-density metrics are

$$\text{median}(D_k(s)), \qquad \frac{\text{sd}(D_k(s))}{\text{mean}(D_k(s))}.$$

Hubness is the Gini coefficient of the in-degree counts h_j. If $h_{(1)} \leq \cdots \leq h_{(n_s)}$, then

$$G_h(s) = \frac{2 \sum_{\ell=1}^{n_s} \ell h_{(\ell)}}{n_s \sum_{\ell=1}^{n_s} h_{(\ell)}} - \frac{n_s + 1}{n_s}.$$

Spectral structure. Let $\lambda_1, \ldots, \lambda_p$ be the positive PCA explained-variance values for the normalized embeddings of subfield s. The dimensionality proxy and normalized spectral entropy are

$$D_{80}(s) = \min\left\{m : \frac{\sum_{j=1}^{m} \lambda_j}{\sum_{j=1}^{p} \lambda_j} \geq 0.80\right\}, \qquad H(s) = -\frac{\sum_{j=1}^{p} p_j \log p_j}{\log p},$$

where $p_j = \lambda_j / \sum_{\ell=1}^{p} \lambda_\ell$. The first quantity counts the PCA components needed to explain 80 percent of variance; the second measures how evenly variance is spread across positive-variance directions.

Separate semantic-displacement indicator. Let c_s^{early} and c_s^{late} be centroids for the early and late windows. In the full thesis period these are 2000–2004 and 2020–2024. Early–late centroid drift is

$$\Delta_c(s) = d(c_s^{early}, c_s^{late}).$$

This quantity is reported as net semantic displacement. It is not included in the eight-metric structural profile, profile PCA, static clustering, or morphological distance matrices.

B. ILLUSTRATIVE UMAP ATLAS OF SELECTED SUBFIELDS

This appendix provides an illustrative atlas of selected subfield-level UMAP projections. The panels are visual diagnostics only: they do not compute morphology metrics, assign clusters, measure distances, or support quantitative claims. Because each panel is a separate subfield-specific projection, axes, distances, shapes, and density scales should not be compared across subfields.

The atlas maps were generated from the row-aligned SPECTER2 analysis matrix using main-analysis records from 2000–2024 and at most 10,000 papers per subfield. The per-subfield UMAP routine (McInnes et al., 2018) used cosine distance with $n_{\text{neighbors}} = 30$, min_dist = 0.05, random_state = 42, and low-memory mode. The density panel uses a smoothed 150-by-150 histogram over displayed UMAP coordinates, clipped at the 99th percentile of positive density values and drawn with `viridis`.

The selected cases connect to recurring empirical themes in the main chapters: atypical profiles, dispersed physical-science structure, compact or separated local organization, computer-science displacement, and broad material-science morphology.

A broader interactive project viewer is available at https://aleetreny.github.io/TFM/. It can be used to explore additional discipline and subfield shapes beyond the selected cases included in this appendix.

Figure B.1

Illustrative UMAP atlas: Human Factors and Ergonomics.

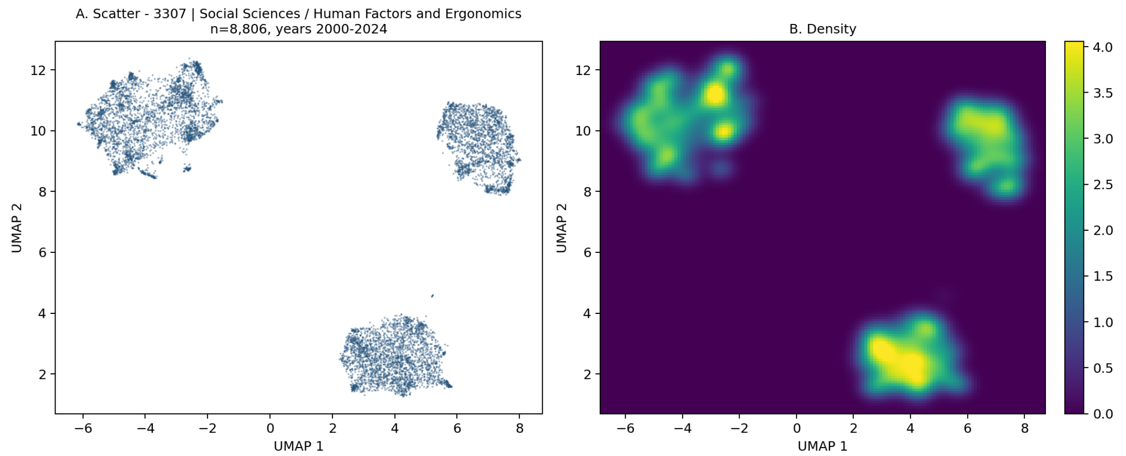

Figure B.2

Illustrative UMAP atlas: Nuclear and High Energy Physics.

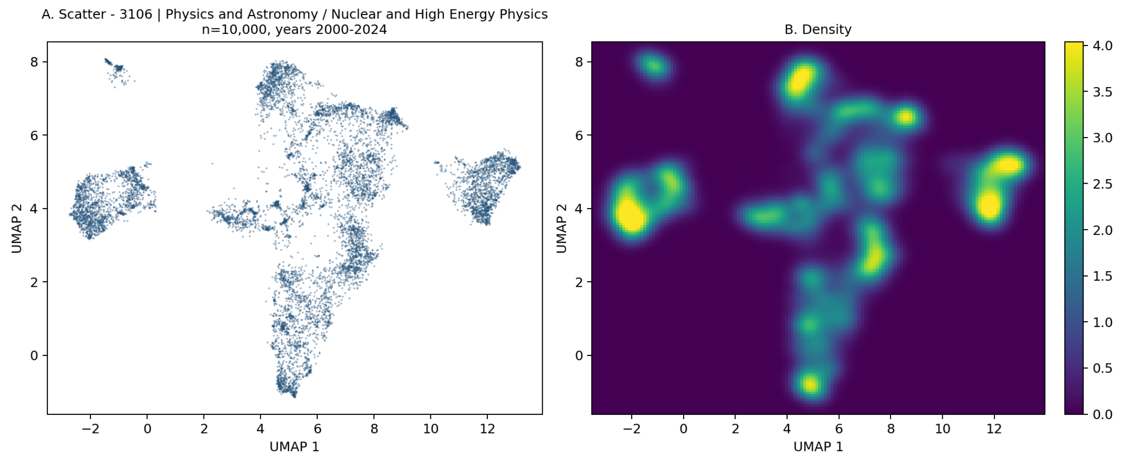

Figure B.3

Illustrative UMAP atlas: Applied Microbiology and Biotechnology.

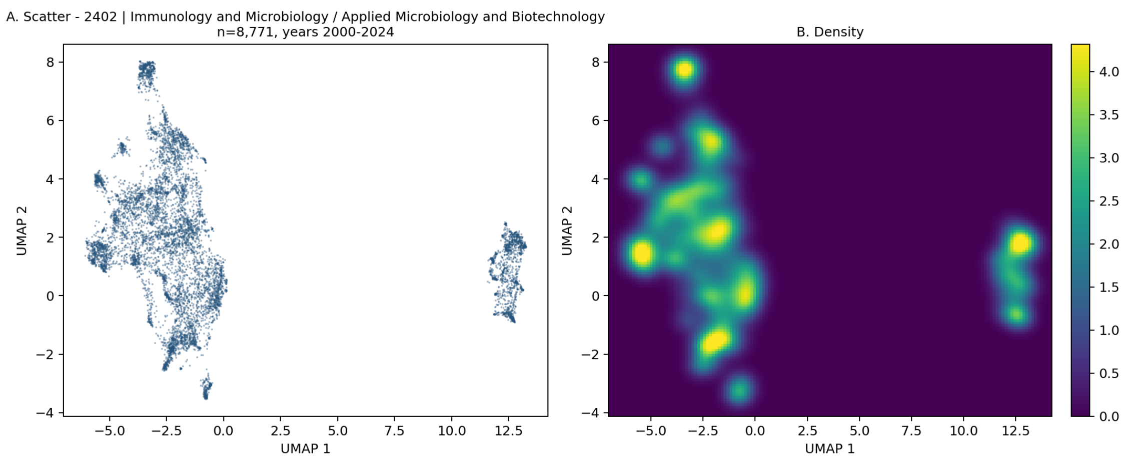

Figure B.4

Illustrative UMAP atlas: Computer Vision and Pattern Recognition.

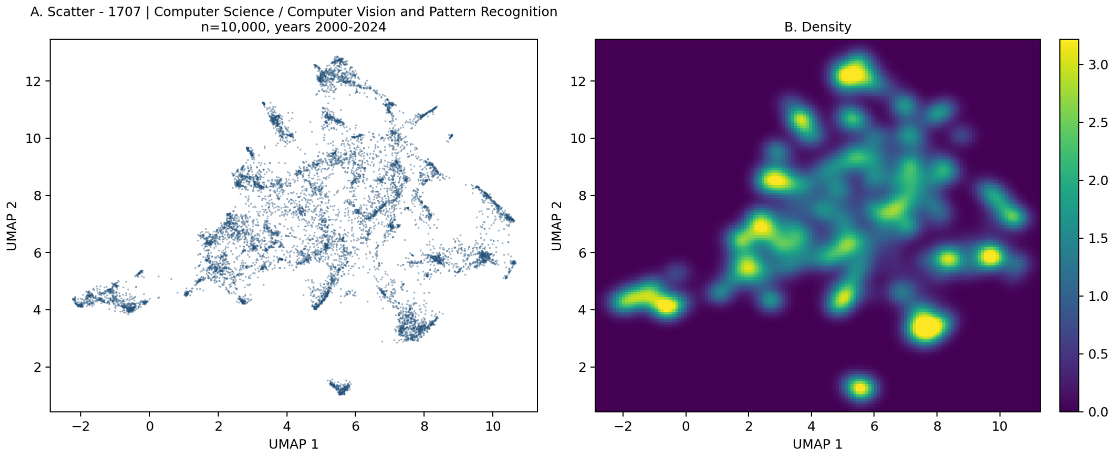

Figure B.5

Illustrative UMAP atlas: Signal Processing.

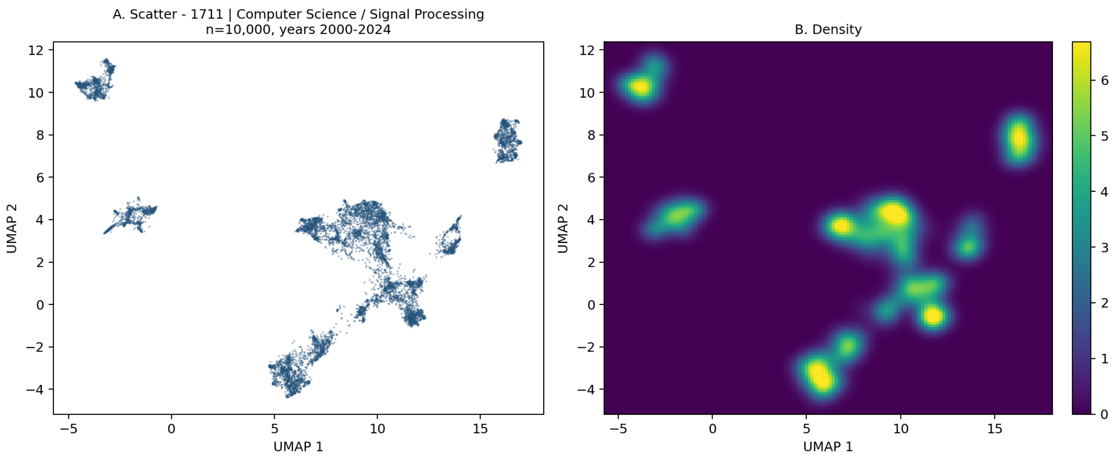

Figure B.6

Illustrative UMAP atlas: General Materials Science.